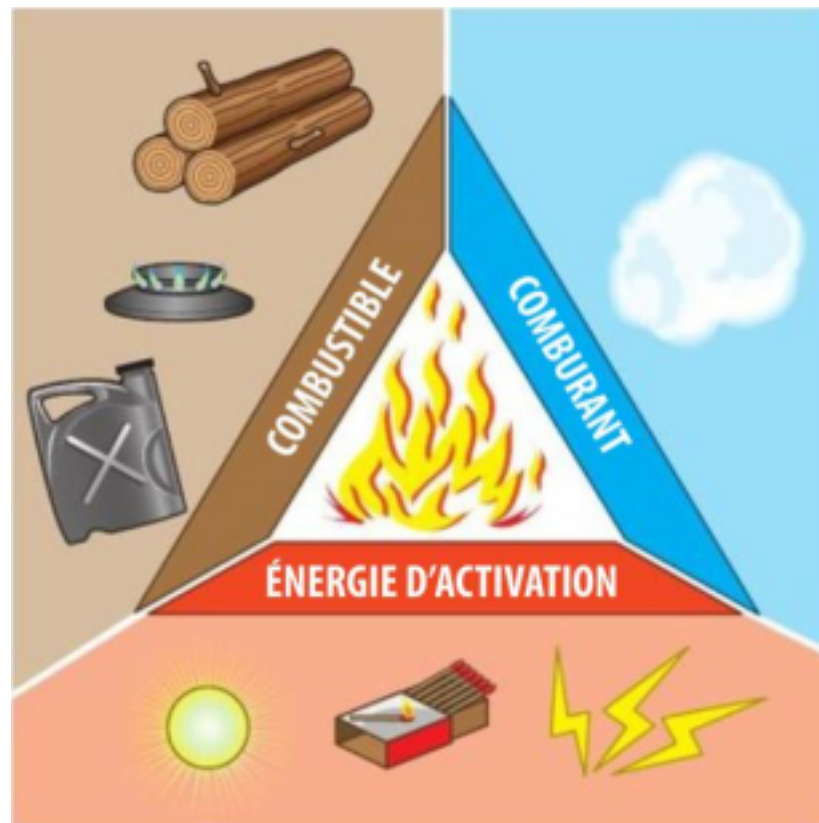
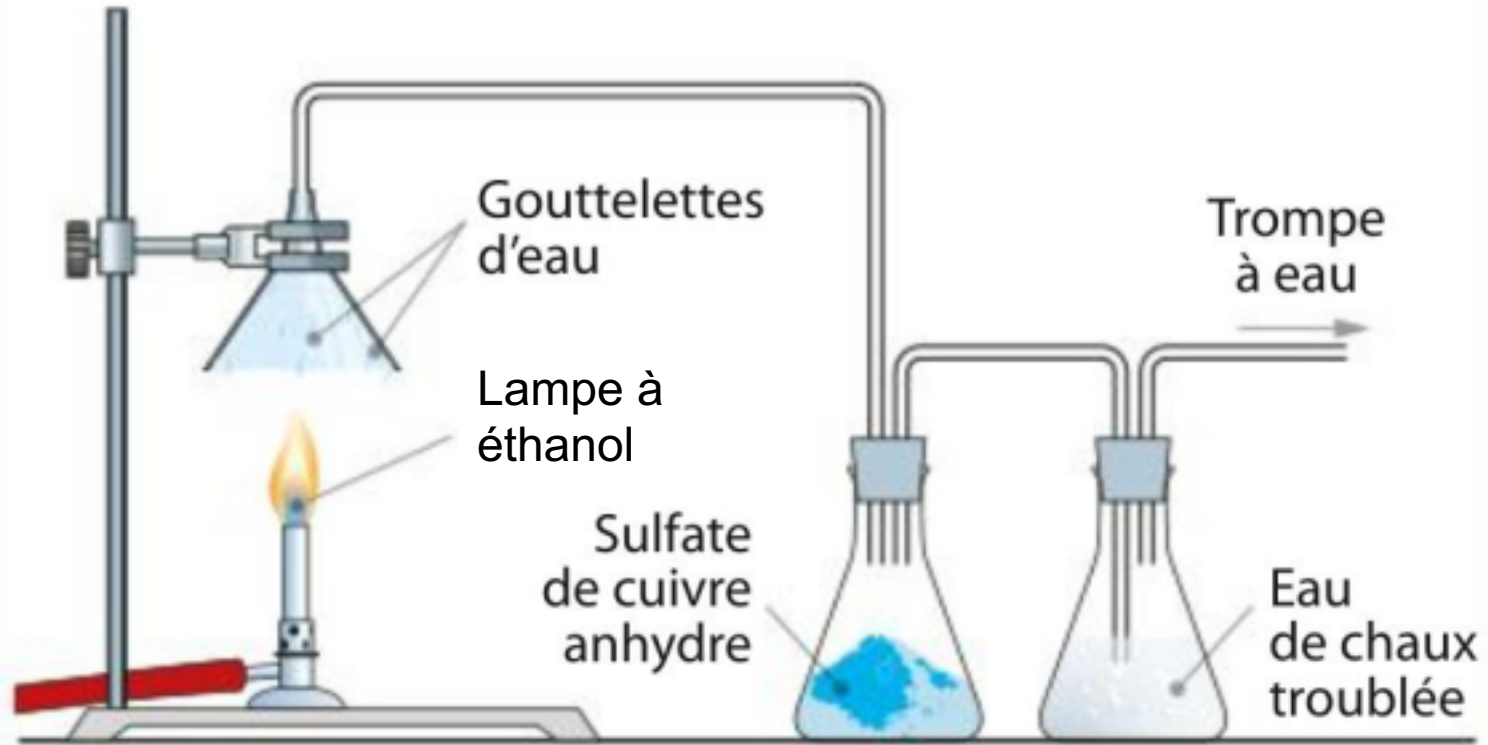


LC02 Énergie chimique

La combustion



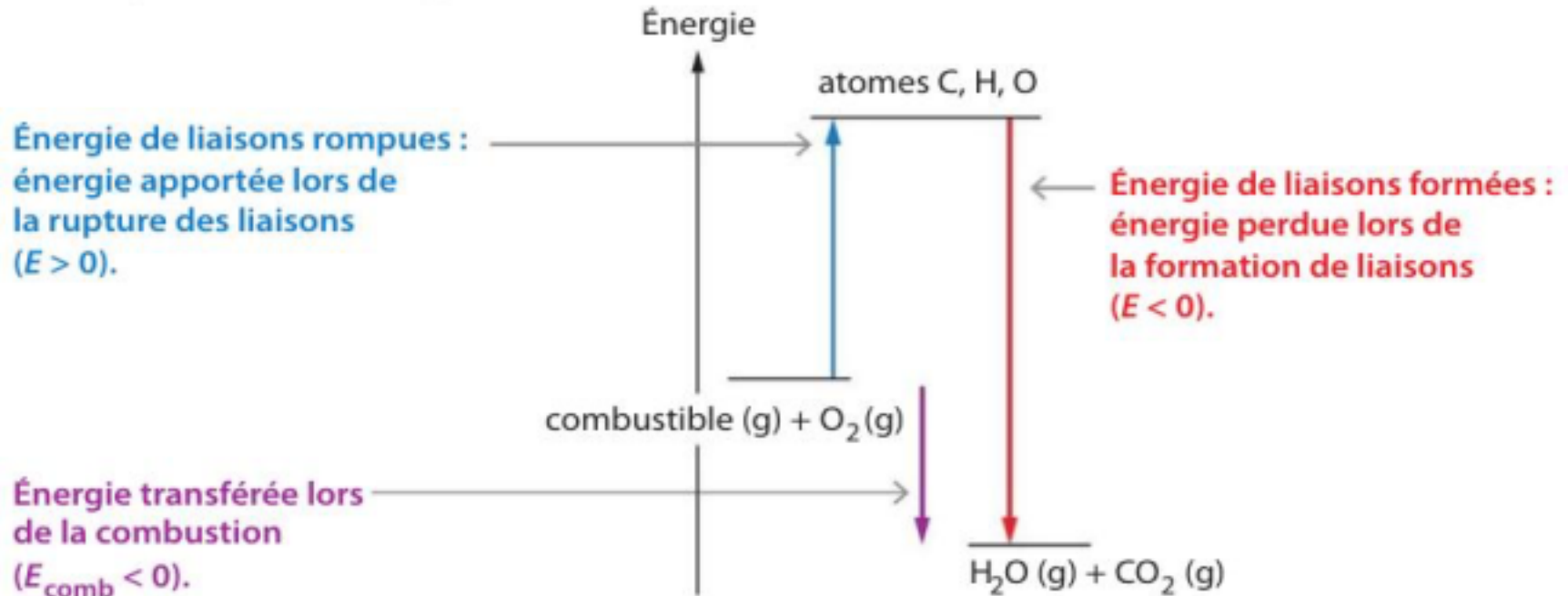
Tests caractéristiques de la combustion de l'éthanol :



Pouvoir calorifique massique de certains combustibles usuels

Combustible	P_c (MJ · kg ⁻¹)
Essence sans plomb	43
Éthanol	29
Gaz naturel	55
Bois	15
Charbon	15-27

Diagramme énergétique de la combustion

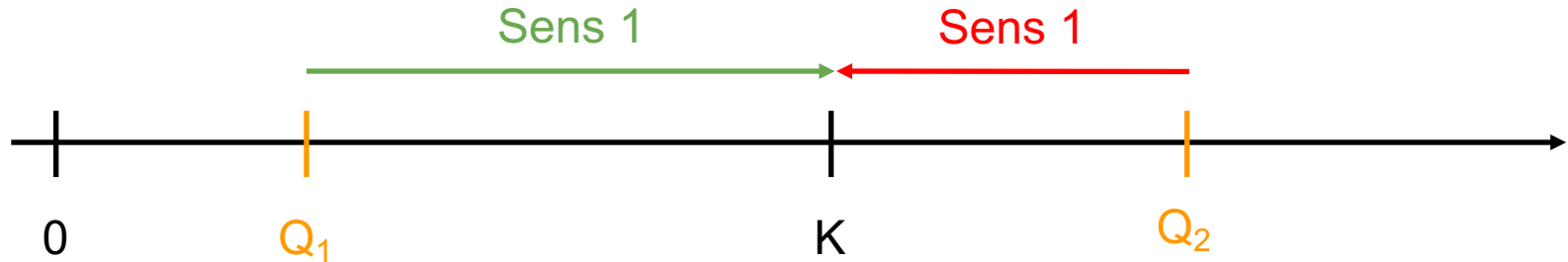


Énergies de liaison

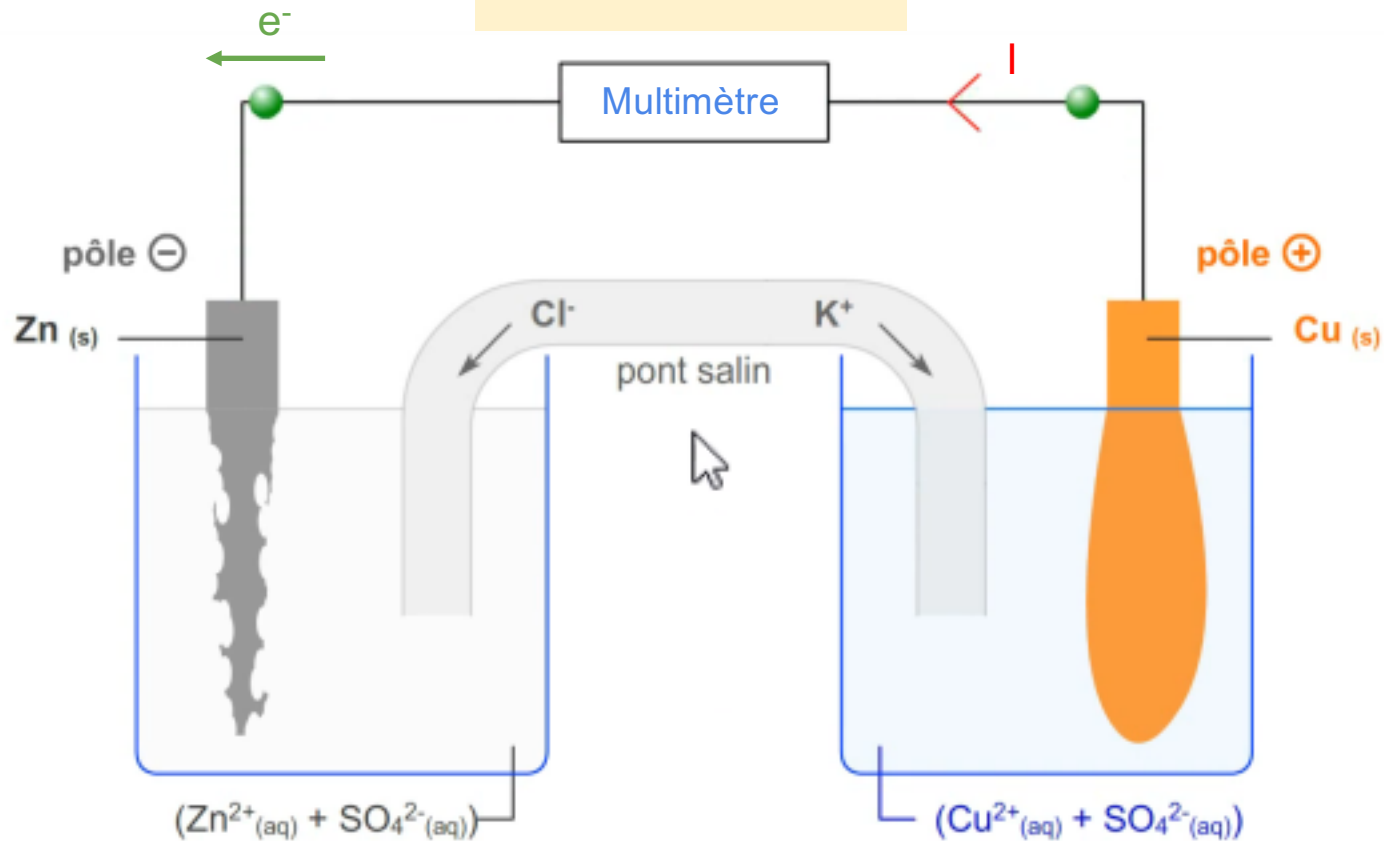
Liaison A-B	$D_{A-B} \text{ (kJ} \cdot \text{mol}^{-1}\text{)}$
C-H	413
C-O	358
O-H	463
C-C	348
C=O	798
O=O	498

Rappel : Critère d'évolution spontané

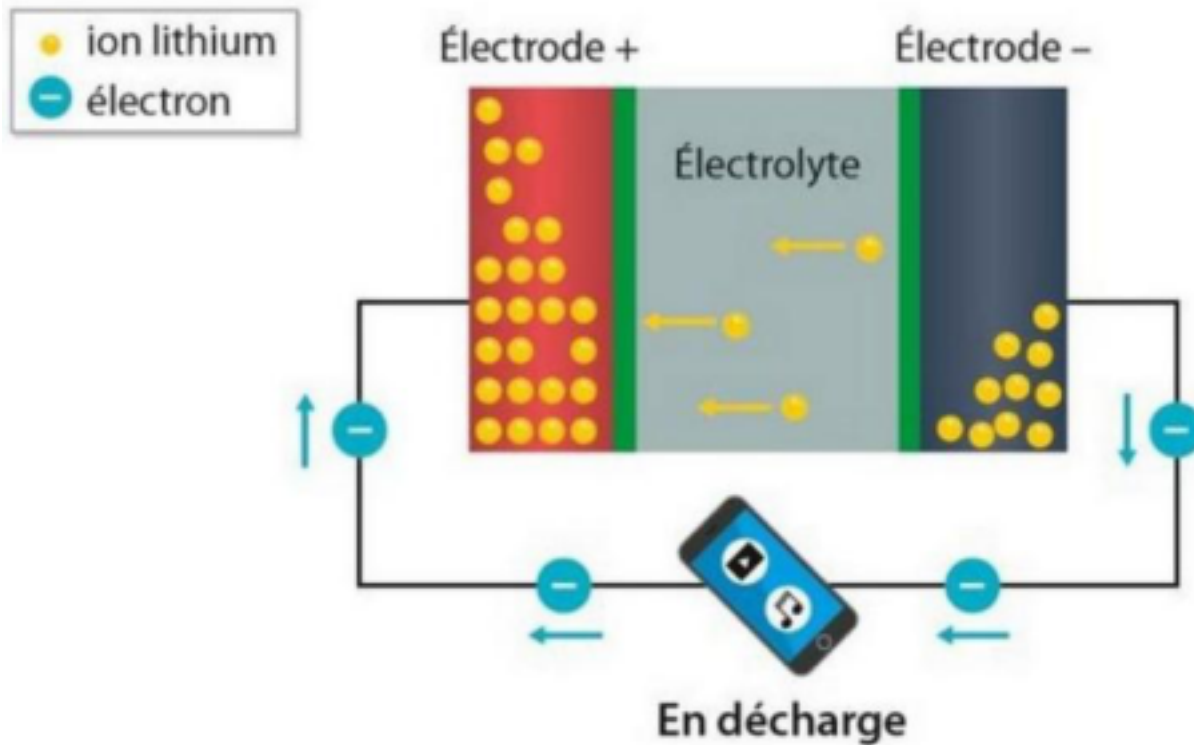
- $Q_1 < K$ □ Sens 1
 Q_1 augmente jusqu'à K .
- $Q_2 < K$ □ Sens 2
 Q_2 diminue jusqu'à K .



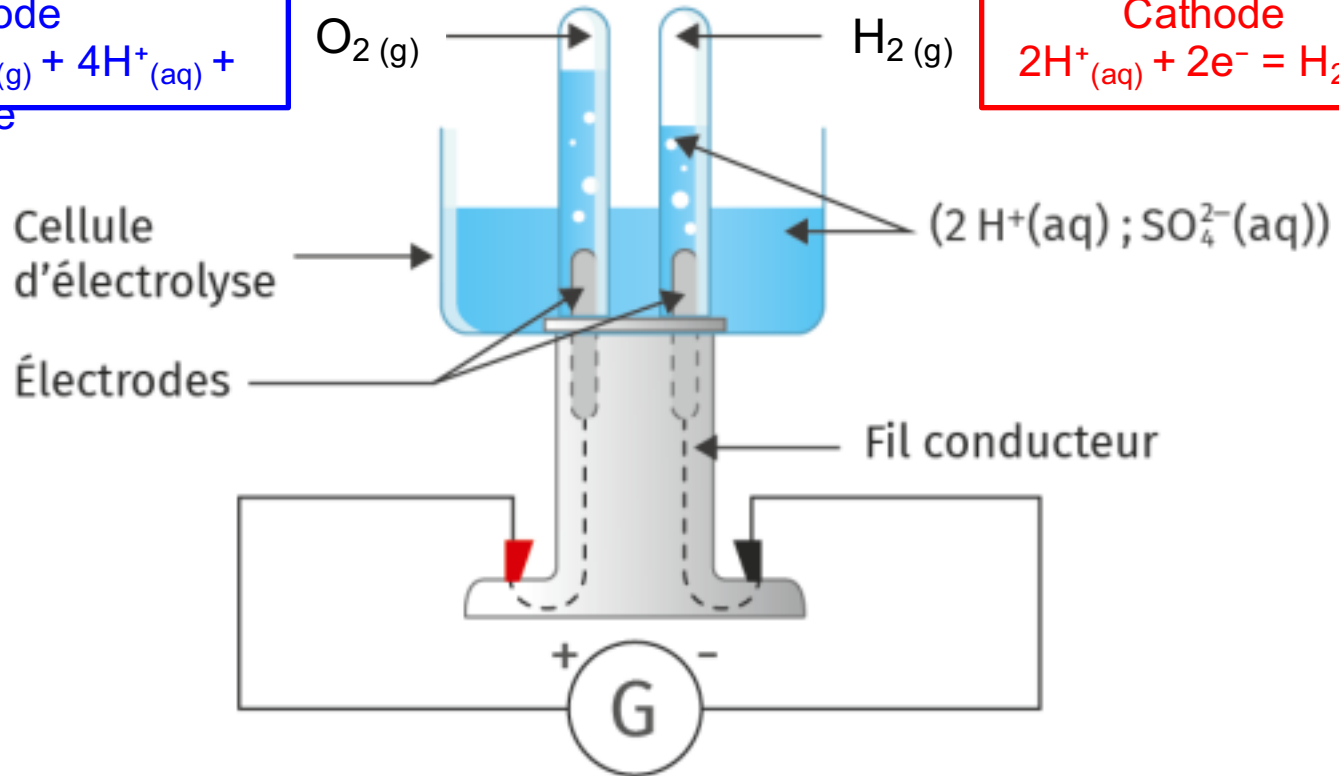
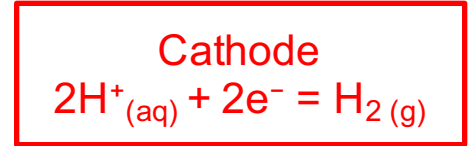
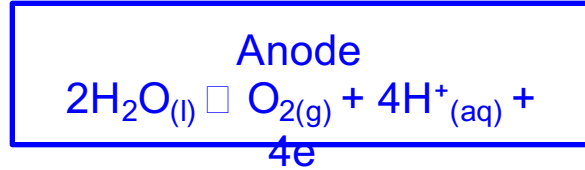
Pile Daniell



Fonctionnement du téléphone sur batterie



Électrolyse de l'eau : $\text{H}_2\text{O}_{(l)} = 2\text{H}_{2(g)} + \text{O}_{2(g)}$



Téléphone en charge

